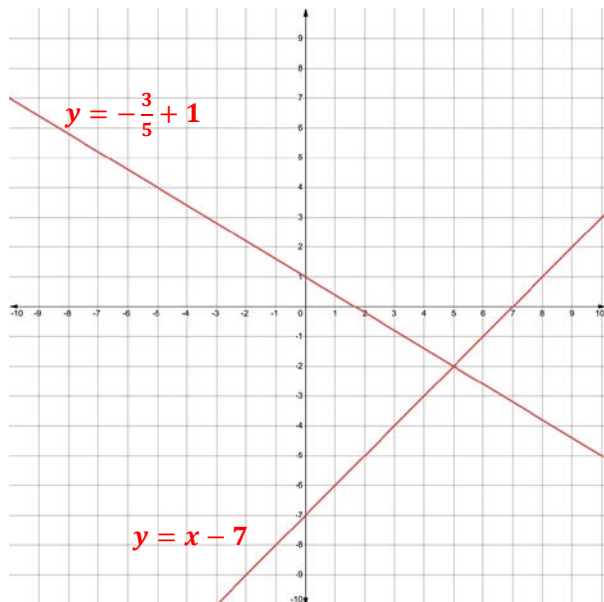


Grade 8
Unit 8
Lesson 9 Practice

Name Answer Key
Date _____
Period _____

Use the coordinate grid to complete questions 1-4.



1. Graph the equation $y = -\frac{3}{5}x + 1$.

Commented [JP1]: <https://www.desmos.com/calculator/bg8y4q5gha>

2. Graph the equation $y = x - 7$.

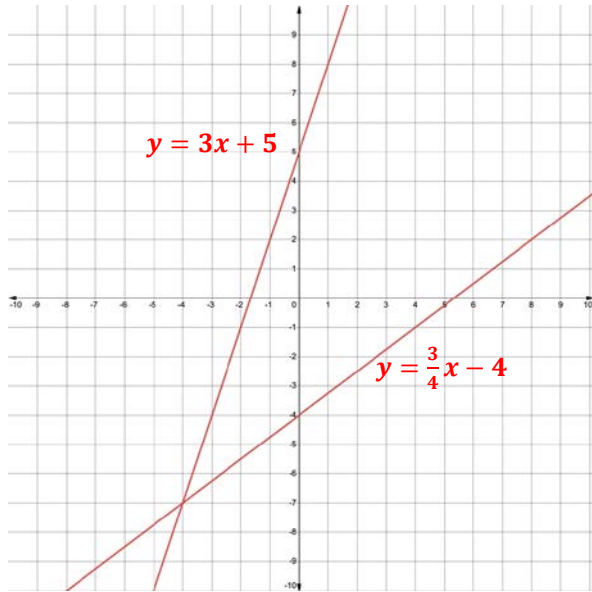
3. Based on your work for questions 1 & 2, what is the solution to the system of equations $\begin{cases} y = -\frac{3}{5}x + 1 \\ y = x - 7 \end{cases}$?

$(5, -2)$

4. Use substitution to show that the solution you identified in question 3 satisfies both equations in the system of equations.

$y = -\frac{3}{5}x + 1$	$y = x - 7$
$-2 = -\frac{3}{5}(5) + 1$ $-2 = -3 + 1$ $-2 = -2 \checkmark$	$-2 = 5 - 7$ $-2 = -2 \checkmark$

Use the coordinate grid to complete questions 5-7.



5. Graph the system of equations

$$\begin{cases} y = \frac{3}{4}x - 4 \\ y = 3x + 5 \end{cases}$$

Commented [JP2]: <https://www.desmos.com/calculator/bg8y4q5ggha>

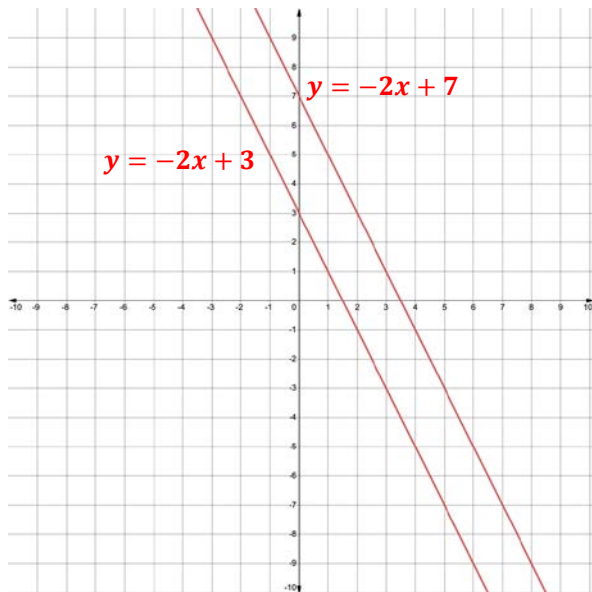
6. What is the solution to the system of equations $\begin{cases} y = \frac{3}{4}x - 4 \\ y = 3x + 5 \end{cases}$?

$(-4, -7)$

7. Use substitution to show that the solution you identified both equations in the system of equations.

$$\begin{array}{ll} y = \frac{3}{4}x - 4 & y = 3x + 5 \\ -7 = \frac{3}{4}(-4) - 4 & -7 = 3(-4) + 5 \\ -7 = -3 - 4 & -7 = -12 + 5 \\ -7 = -7 \checkmark & -7 = -7 \checkmark \end{array}$$

Use the coordinate grid to complete questions 8-10.



8. Graph the system of equations

$$\begin{cases} y = -2x + 3 \\ y = -2x + 7 \end{cases}$$

Commented [JP3]: <https://www.desmos.com/calculator/bg8y4q5ggha>

9. Complete the statement: The two lines in the system of equations

☒ are parallel
☐ intersect
☐ coincide

10. Describe the solution set to the system of equations $\begin{cases} y = -2x + 3 \\ y = -2x + 7 \end{cases}$

There are no solutions.